

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604I - NUMERICAL METHODS IN CIVIL ENGG.

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following methods is direct in nature for solving simultaneous linear equations?
 - A) Gauss-Seidel Method
 - B) Jacobi Iteration
 - C) Gauss Elimination Method
 - D) Successive Over Relaxation
 - 2 Gauss elimination method is mainly used for:
 - A) Finding eigenvalues
 - B) Solving linear simultaneous equations
 - C) Root finding of nonlinear equations
 - D) Numerical integration
 - 3 Cholesky decomposition is applicable when the coefficient matrix is:
 - A) Singular and rectangular
 - B) Diagonal and sparse
 - C) Symmetric and positive definite
 - D) Upper triangular
 - 4 The trial and error method for root finding is also known as:
 - A) Bisection method
 - B) Bracketing method
 - C) Incremental search method
 - D) Secant method
 - 5 Which method always guarantees convergence if the function changes sign within the interval?
 - A) Newton–Raphson method
 - B) Secant method
 - C) Bisection method
 - D) Iteration method

- 6 Newton–Raphson method may fail if:
 - A) Initial guess is close to root
 - B) Derivative of function is zero or very small
 - C) Function is continuous
 - D) Interval is small
- 7 Newton’s divided difference interpolation is preferred when:
 - A) Data points are equally spaced
 - B) Data points are unequally spaced
 - C) Only two data points are known
 - D) Errors are large
- 8 Lagrange’s interpolation formula is advantageous because it:
 - A) Requires finite differences
 - B) Does not require tabular differences
 - C) Is iterative
 - D) Requires matrix inversion
- 9 Trapezoidal rule assumes the curve between two points to be:
 - A) A straight line
 - B) A parabola
 - C) A cubic polynomial
 - D) A sine curve
- 10 Simpson’s 1/3 rule requires the number of intervals to be:
 - A) Odd
 - B) Prime
 - C) Even
 - D) Multiple of three
- 11 Which measure best represents dispersion of data?
 - A) Mean
 - B) Median
 - C) Variance
 - D) Mode
- 12 Least square method aims to:
 - A) Minimize absolute errors
 - B) Minimize square of errors
 - C) Eliminate experimental errors
 - D) Maximize correlation

Q2. Solve the following.

- A) Solve the following system using Gauss Elimination: 6

$$\begin{aligned} 2x + 3y + z &= 1 \\ 4x + y + 2z &= 2 \\ 3x + 2y + 3z &= 3 \end{aligned}$$
- B) What is Cholesky Decomposition? When is it used? 6

Q3. Solve the following.

- A) Explain the Bisection Method for finding the root of an equation. Apply the method to find the root of the equation 6

$$f(x) = x^3 - x - 2$$
 in the interval $[1, 2]$ up to 3 iterations.
- B) What is the Newton-Raphson Method? Solve $f(x) = x^2 - 5$ with initial guess $x_0=2$, 6
 Perform 2 iterations.

Q4. Solve Any Two of the following.

- A) Use the Finite Difference Method (Central Difference) to approximate the first derivative of the function $f(x)=x^2 + 3x + 2$ at $x = 1$ with a step size of $h = 0.1$.
- B) Find the value of $f(2.5)$ using Lagrange's Interpolation Formula from the given data:

x	F(x)
2	1
3	2
4	5

- C) Use Newton's Divided Differences Method to find the value of $f(3.5)$ for the following data points:

x	F(x)
3	5
3.2	5.6
3.4	6.2

Q5. Solve Any Two of the following.

- A) Use Simpson's $\frac{1}{3}$ Rule to approximate the integral of $f(x)=x^2$ from $x=1$ to $x=3$ using two intervals. 6
- B) Use Gaussian Quadrature (2-point) to approximate the integral of $f(x) = e^x$ from $x=0$ to $x=1$. 6
- C) Use the Trapezoidal Rule to approximate the integral of $f(x)=x^2$ from $x=1$ to $x=3$ using two intervals. 6

Q6. Solve Any Two of the following.

- A) What is the difference between mean, median, and mode? Explain with examples. 6
- B) Fit the Parabola of the type $y = a + bx + cx^2$ for the following data: 6

x	20	30	40	50	60
y	54	90	138	206	292

- C) Fit the Straight line to the following data 6

x	1	2	3	4	6	8
y	2.4	3	3.6	4	5	6

*** End ***